Contents

0.1 Cantor Set

Define operations on $\mathcal{P}(\mathbb{R})$ as: for $A, B \subset \mathbb{R}$ and $c \in \mathbb{R}$,

$$A + B \stackrel{\text{def}}{=} \{a + b \mid a \in A, b \in B\}$$
$$cA \stackrel{\text{def}}{=} \{ca \mid a \in A\}$$

If $A = \{\alpha\}$ is a singleton, denote $A + B$ as $\alpha + B$.

Definition 0.1.1. Let $C_0 \stackrel{\text{def}}{=} [0, 1]$ and $C_n \stackrel{\text{def}}{=} \frac{C_{n-1}}{3} \cup \left(\frac{2}{3} + \frac{C_{n-1}}{3}\right)$, for $n \geq 1$.

An intersection $C = \bigcap_{n=1}^{\infty} C_n$ is called a *Cantor Set*.

Cantor set is a remarkable object in the Real numbers set.
In particularly, Cantor Set is *Uncountable*, *Compact*, *Perfect*, *Nowhere Dense*, and has *Measure zero*.
Moreover, sum of two Cantor sets is $[0, 2]$.
And, total length (or measure) of C is zero, because:

$\#C_0 = 1,$

$\#C_1 = \frac{2}{3} = 1 - \frac{1}{3},$

$\#C_2 = 1 - \frac{1}{3} - \frac{1}{3^2} \cdot 2^1$

$\vdots$

$$\#C_n = 1 - \sum_{i=0}^{n} \frac{2^i}{3^{i+1}} = 1 - \frac{1}{3} \cdot \frac{1 - \left(\frac{2}{3}\right)^n}{1 - \frac{2}{3}} = 1 - \left(1 - \left(\frac{2}{3}\right)^n\right) = \left(\frac{2}{3}\right)^n.$$

0.2 Ternary Expansion

Definition 0.2.1. Given $x \in (0,1)$, a sequence $\{n_i\}_{i=1}^{\infty}$ in $\{0,1,2\}$ is called a *ternary expansion* of x if:

$$x = \sum_{i=1}^{\infty} \frac{n_i}{3^i}$$

Theorem 0.2.2. Given $x \in (0,1)$, a ternary expansion of x exists.

Proof. Let $x \in (0,1)$ be given. We construct the sequence $\{n_i\}$ inductively as follows:
First, choose $n_1 \in \{0,1,2\}$ to be the largest integer such that

$$\frac{n_1}{3} < x$$

Next, for chosen $n_1, n_2, \ldots, n_{k-1}$, we choose the largest digit $n_k \in \{0,1,2\}$ such that

$$\frac{n_1}{3} + \frac{n_2}{3^2} + \cdots + \frac{n_{k-1}}{3^{k-1}} + \frac{n_k}{3^k} < x$$

From this construction of $\{n_i\}$, it follows that for any $k \in \mathbb{N}$,

$$\sum_{i=1}^{k} \frac{n_i}{3^i} < x \leq \sum_{i=1}^{k} \frac{n_i}{3^i} + \frac{1}{3^k}$$

Note that the partial sum $\sum_{i=1}^{k} \frac{n_i}{3^i}$ increases monotonically and is bounded by 1, thus it converges.
Taking the limit as $k \to \infty$ in the above inequality, we obtain:

$$\sum_{i=1}^{\infty} \frac{n_i}{3^i} \leq x \leq \sum_{i=1}^{\infty} \frac{n_i}{3^i} + \lim_{k\to\infty} \frac{1}{3^k}$$

Since $\lim_{k\to\infty} \frac{1}{3^k} = 0$, we have

$$x = \sum_{i=1}^{\infty} \frac{n_i}{3^i}$$

which gives the desired result. $\square$

Note: the ternary expansion described in the existence theorem is called the *First-form* ternary expansion.

> **Theorem 0.2.3.** Given $x \in (0,1)$, the ternary expansion of x is unique except: for some $m \in \mathbb{N}$ and $0 < a < 3^m$,
>
> $$x = \frac{a}{3^m}, \text{ irreducible form} \tag{*}$$
>
> Particularly, if x can be represented as $(*)$, it has only two different ternary expansion.

Proof. First, we show that if x is of the form $(*)$, it has only two expansions. If x can be represented as $(*)$, then

$$x = \frac{a}{3^m} = \frac{a_1}{3} + \frac{a_2}{3^2} + \cdots + \frac{a_m}{3^m}$$

where $a_i \in \{0,1,2\}$ with $a_m \neq 0$. It is obtained by the Euclidean Algorithm:

$$a = 3^{m-1}a_1 + 3^{m-2}a_2 + \cdots + 3^2 a_{m-2} + 3a_{m-1} + a_m$$

(If $a_m = 0$, then x can be divided by 3, it is contradiction to irreducibility.) Meanwhile, note that

$$\sum_{k=m+1}^{\infty} \frac{2}{3^k} = \frac{2}{3^{m+1}} \cdot \frac{1}{1-1/3} = \frac{2}{3^{m+1}} \cdot \frac{3}{2} = \frac{1}{3^m}.$$

Therefore, we obtain two ternary expansion of x as:

$$x = \frac{a_1}{3} + \frac{a_2}{3^2} + \cdots + \frac{a_m}{3^m} + \sum_{k=m+1}^{\infty} \frac{0}{3^k} = \frac{a_1}{3} + \frac{a_2}{3^2} + \cdots + \frac{a_m - 1}{3^m} + \sum_{k=m+1}^{\infty} \frac{2}{3^k} \tag{**}$$

This proves that such x has two distinct expansions. Meanwhile, suppose that x has two distinct expansions $\{n_i\}$, $\{r_i\}$. Let $j \in \mathbb{N}$ be the smallest index such that $n_j \neq r_j$. Since $n_i = r_i$ for all $i < j$, we have:

$$0 = \sum_{i=1}^{\infty} \frac{n_i - r_i}{3^i} = \frac{n_j - r_j}{3^j} + \sum_{i=j+1}^{\infty} \frac{n_i - r_i}{3^i} \tag{***}$$

Since $n_j \neq r_j$ and

$$\left| \sum_{i=j+1}^{\infty} \frac{n_i - r_i}{3^i} \right| \leq \sum_{i=j+1}^{\infty} \frac{2}{3^i} = \frac{1}{3^j}$$

the $(***)$ holds *if and only if*

$$n_j - r_j = 1,\ n_i - r_i = -2 \text{ for all } i \geq j+1 \text{ or } n_j - r_j = -1,\ n_i - r_i = 2 \text{ for all } i \geq j+1$$

But this equivalent to x can be represented in the form of $(*)$, therefore we proved the Uniqueness. $\square$

Note: the ternary expansion is of the form $(*)$ has two expansions. Clearly, the left side in $(**)$ is a first form. The right side form is called a *Second form*.

0.3 Ternary representation of Cantor Set

Theorem 0.3.1. $C = \left\{ \sum_{i=1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\} \subseteq \{0, 2\} \right\}$ where C is the Cantor set.

Proof. Since the existence of ternary expansion,

$$C_0 = [0,1] = \left\{ \sum_{i=1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\} \subseteq \{0,1,2\} \right\}$$

Observe that

$$\begin{aligned}
C_1 &= \frac{C_0}{3} \cup \left(\frac{2}{3} + \frac{C_0}{3} \right) = \left\{ \frac{0}{3^1} + \sum_{i=2}^{\infty} \frac{n_i}{3^i} \mid \{n_i\}_{i=2}^{\infty} \subseteq \{0,1,2\} \right\} \cup \left\{ \frac{2}{3^1} + \sum_{i=2}^{\infty} \frac{n_i}{3^i} \mid \{n_i\}_{i=2}^{\infty} \subseteq \{0,1,2\} \right\} \\
&\vdots \\
C_k &= \bigcup_{n_1,\ldots,n_k \in \{0,2\}} \left\{ \frac{n_1}{3^1} + \frac{n_2}{3^2} + \cdots + \frac{n_k}{3^k} + \sum_{i=k+1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\}_{i=k+1}^{\infty} \subseteq \{0,1,2\} \right\} \\
&= \left\{ \sum_{i=1}^{k} \frac{n_i}{3^i} + \sum_{i=k+1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\}_{i=1}^{k} \subseteq \{0,2\} \text{ and } \{n_i\}_{i=k+1}^{\infty} \subseteq \{0,1,2\} \right\}
\end{aligned}$$

Finally,

$$C \overset{\text{def}}{=} \bigcap_{k \in \mathbb{N}} C_k = \left\{ \sum_{i=1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\} \subseteq \{0,2\} \right\}$$

□

Theorem 0.3.2. $C + C = [0,2]$ where C is the Cantor set.

Proof. Since $C \subseteq [0,1]$, $C + C \subseteq [0,2]$ is clear. Let $x \in [0,2]$ be given.
Trivially $\frac{x}{2} \in [0,1]$. Therefore, there exists a ternary expansion $\{n_i\} \subseteq \{0,1,2\}$ s.t.

$$\sum_{i=1}^{\infty} \frac{n_i}{3^i} = \frac{x}{2}$$

Use this fact, we can write that

$$x = \sum_{i=1}^{\infty} \frac{2n_i}{3^i}$$

Define

$$a_i = \begin{cases} 0 & \text{if } n_i = 0 \\ 0 & \text{if } n_i = 1 \\ 2 & \text{if } n_i = 2 \end{cases} \quad \text{and} \quad b_i = \begin{cases} 0 & \text{if } n_i = 0 \\ 2 & \text{if } n_i = 1 \\ 2 & \text{if } n_i = 2 \end{cases}$$

Then, $2n_i = a_i + b_i$ for all $i \geq 1$. And,

$$x = \sum_{i=1}^{\infty} \frac{2n_i}{3^i} = \sum_{i=1}^{\infty} \frac{a_i}{3^i} + \sum_{i=1}^{\infty} \frac{b_i}{3^i} \in C + C$$

□

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Prop. Let C be a Cantor set, and χ_C be a Characteristic function defined on $[0,1]$

1). The map χ_C is Continuous on $[0,1]\setminus C$.

2). The map χ_C is discontinuous at every point on C.

3). The map χ_C is differentiable on $[0,1]\setminus C$.

4). The map χ_C is Riemann-Integrable on $[0,1]$.

Proof. All C_n $(n\in\mathbb{N})$ are closed, so $C=\bigcap_{n\in\mathbb{N}} C_n$ is also closed set.
Therefore, given $x\in[0,1]\setminus C$, there exists $r>0$ such that $(x-r, x+r)\subseteq[0,1]\setminus C$.
Now, for all $t\in(x-r,x+r)$, $|\chi_C(x)-\chi_C(t)|=|0-0|=0$, so χ_C is Continuous.
Similarly, 3) Can be proved. To show the second assertion, let $x\in C$ be given
Fix $\varepsilon=\frac{1}{2}$. Given $\delta>0$, choose $n\in\mathbb{N}$ such that $1/3^n<\delta$. Denote $B=(x-\delta,x+\delta)\cap[0,1]$.
By definition, $x\in C_n$ and there exists $t\in B\setminus C_n$, because:
if no such t exists, that is, $B\setminus C_n=\emptyset$, it equals to $B\subseteq C_n$.
But, the C_n is union of disjoint closed intervals of length $1/3^n$, it contradicts $1/3^n<\delta$.
Thus, $C\subseteq C_n$ implies $t\in B\setminus C_n\subseteq B\setminus C$, it implies

$$|f(x)-f(t)|=1>\frac{1}{2}=\varepsilon.$$

Consequently, χ_C is not Continuous at $x\in C$.
To show the last assertion, observe that

$$0\le\underline{\int_0^1}\chi_C(x)\,dx\le\overline{\int_0^1}\chi_C(x)\,dx\le\left(\frac{2}{3}\right)^n\cdot 3,$$ because:

Let $P_n:=\{a/3^n \mid a\in\mathbb{Z},\ 0\le a\le 3^n\}$ be a partition on $[0,1]$, then the upper sum

$$U(f,P_n)=\sum_{i=1}^{3^n} M_i\cdot\frac{1}{3^n} \quad \text{where } M_i=\sup\{\chi_C(x) \mid \tfrac{i-1}{3^n}\le x\le\tfrac{i}{3^n}\}$$

$$\le\frac{2^n\cdot 3}{3^n}$$

so, $\overline{\int_0^1}\chi_C\le U(f,P_n)$ by the definition, we obtain the inequality.
Consequently, letting $n\to\infty$, we obtain the result. ▨

Prop. Let C be the Cantor set, and let f be a continuous at except C.
Then, f is Riemann-integrable.